

FIGURE 1
BioPLEASE execution pipeline and data flow

v2.0.0 · local-first
runner.ts → OpenClaude CLI
stream-json · MCP

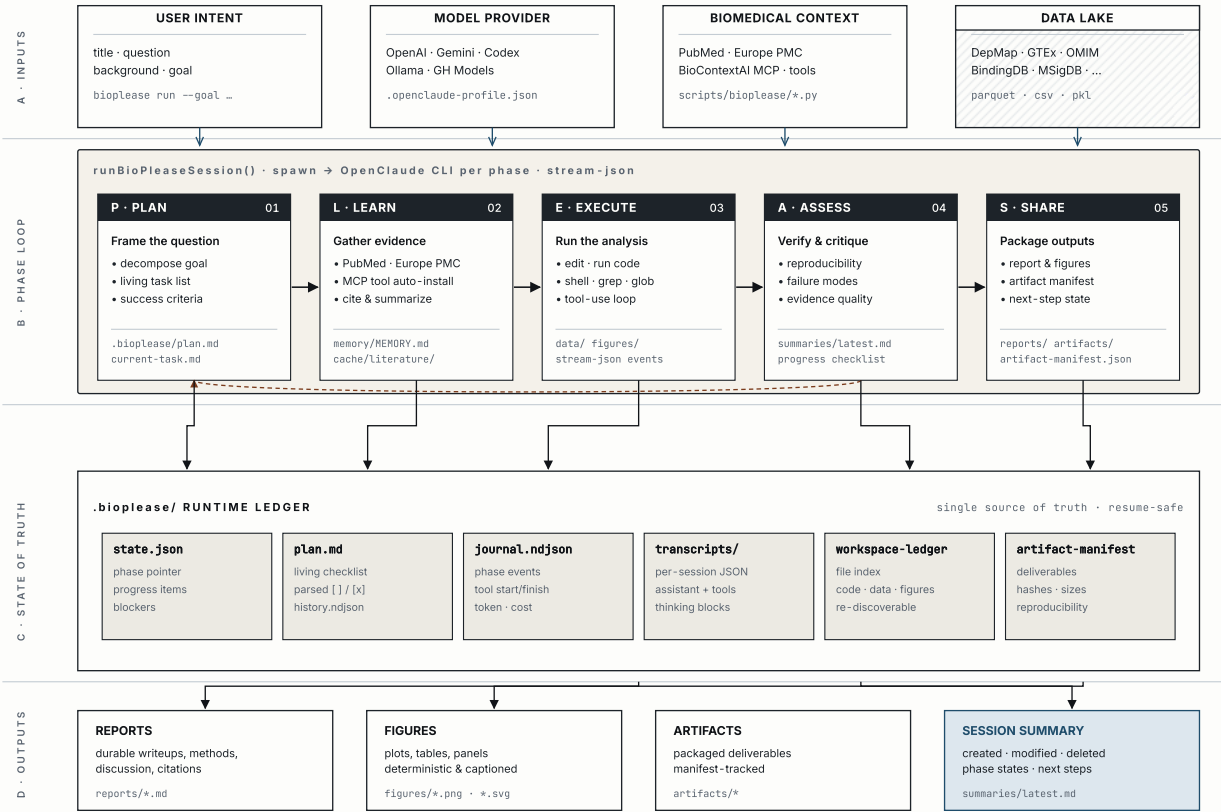


Figure 1. Overview of the BioPLEASE execution pipeline. A user goal, a configured model provider, biomedical retrieval tooling, and a curated local data lake (A) are consumed by a five-phase loop — Plan, Learn, Execute, Assess, Share (B) — orchestrated by `runBioPleaseSession()`, which spawns a fresh OpenClaude CLI subprocess per phase and consumes its `stream-json` output.

Every phase reads from and writes to the `.bioplease/` runtime ledger (C): a phase-pointer state file, a living plan, an append-only journal, per-session transcripts, a workspace file index, and an artifact manifest. This ledger is the single source of truth and makes runs resumable, auditable, and reproducible.

The Assess → Plan feedback edge captures re-planning on new evidence, graceful-stop handling, and failure recovery: the runner finishes the current phase, then jumps back to Plan / Share before stopping. Final deliverables (D) — reports, figures, packaged artifacts, and a session summary — are emitted by Share and indexed in the artifact manifest.

- phase tile (header bar = phase ID)
- ledger artifact in `.bioplease/`
- human-facing output
- input / external resource
- feedback / re-plan edge